

The same nine readings, scored honestly

We are taking the readings measureQuick takes and refusing to dock a customer for owning a 14 SEER system. The protocol drafts below are mine and thinner than they should be. Answer what you know, skip what you do not, and use the open boxes — disagreeing with me is the point. Kitchen and laundry are on their own sheet.

Three ways to answer this, all equally good

Type into it and email the file back · print it and write on it · hand it to Cayden and talk it through
Answer what you know and skip the rest. A page with four answers on it is worth more than a blank one.

Who is answering

So the answers can be attributed in the meeting. Nothing else is collected.

YOUR NAME

ROLE

How the score works, and why

We take the same nine readings measureQuick takes. What we do with them is deliberately different, and that difference is the first thing to check with you.

1 **Efficiency rating is reported, never scored. Age now counts for 25%.**

A 14 SEER system in good condition scores like a system in good condition — nothing in the score moves because a system is not 20 SEER. Age used to work the same way, and that has changed: measured performance is 75% and age is 25%, because a fifteen-year-old system that meets its nameplate is not really a 100% system. At that age repeat failures start, and a score that ignores it sets the customer up to be nickel-and-dimed. Both parts are printed separately, so an older well-kept system can be seen scoring full marks on everything we measured.

scored ·
measured performance 75% (capacity, airflow, charge, static, electrical) + age 25% reported only ·
efficiency rating

The expected-life figures behind the age term are no longer my invention: ASHRAE puts a residential split system at a 15-year median, DOE/AHRI agrees, and Carrier states 15–20 years with annual maintenance against 10–12 without — which is roughly the spread between this table's mass and luxury columns. Ductless equipment appears in none of those sources and is still a Wilson estimate, flagged as one.

PICK ONE

- ☐ Right — 75% measured, 25% age, efficiency reported only
- ☐ Age should count for less than 25%
- ☐ Age should not count at all — go back to condition only
- ☐ Age should count for more than 25%

What a customer with an older, well-kept system should read on their report

2 **Are these the right weights?**

Five scored dimensions, and I picked the split. Capacity carries the most because it is the closest thing to “is it doing the job”, but that is an argument, not a fact.

capacity 30% · airflow 20% · charge 20% · static 15% · electrical 15%

There is also a floor: if fewer than 60% of the readings a dimension needs were taken, the system gets no score at all rather than a flattering one. An earlier version happily scored a system 100 on 40% of the data.

PICK ONE

- ☐ The weights are about right
- ☐ Airflow should carry more
- ☐ Charge should carry more
- ☐ Change them — see my notes

The weighting you would use, and why

3 When the tool should say something about replacement

The guardrail as built: no report recommends replacement, no dollar figure is ever invented, and the score never moves because of age. What a report can say is that a system is at 85% of expected life with two dimensions out of band — facts, in the customer's hands, early enough to plan a spring change-out instead of a July emergency.

PICK ONE

- ☐ That is right — facts only, never a recommendation
- ☐ Go further — let the report say “plan to replace” when the readings justify it
- ☐ Pull back — do not mention remaining life at all

Where you would draw that line, and the words you would use with a customer

The bands that are still blank

This is the part that actually holds the HVAC side up. Every reading without a band is a number nobody can judge, and one of them is the reason refrigerant charge is still a technician's rating instead of a measurement.

4 Refrigerant charge

Charge is the one scored dimension that is still your judgement rather than a reading, and only because these bands are unset. Fill them in and charge becomes 20% of the score computed from superheat and subcooling.

Superheat, subcooling, approach, delta-T, total external static and amps-against-RLA are all computed from what you enter. Nothing is guessed: there is no refrigerant saturation table in the tool, so you enter saturation temperature as the gauge reads it — dew point suction side, bubble point liquid side — or the derived value is simply not shown.

SUPERHEAT, TXV / EEV e.g. 8 - 14 F

SUPERHEAT, FIXED ORIFICE target chart, or a band

SUBCOOLING e.g. 8 - 14 F

APPROACH, IF WE BAND IT liquid line above ambient

CONDITIONS THESE ASSUME e.g. above 75 F outdoor, 20 min runtime

DIFFERENT BY REFRIGERANT? R-410A vs R-454B vs R-22

How you would want the app to behave when conditions are wrong for a charge check

5 **Airflow and static pressure**

The draft is 350–450 CFM per ton, which is the textbook range and probably too wide to be useful in August. Total external static is judged against the blower's nameplate maximum rather than a rule of thumb, which is right in principle and needs a sanity check from someone who has actually measured a few hundred of them.

draft · 350 - 450 CFM per ton · static judged against nameplate max ESP

CFM PER TON, COOLING e.g. 380 - 420

TEMPERATURE RISE, HEATING e.g. nameplate rise band

DELTA-T ACROSS THE COIL e.g. 16 - 22 F at 50% RH

STATIC: NAMEPLATE, OR A FIXED CEILING? e.g. 0.5 in. w.c. regardless

What you actually do when a system cannot make airflow — and what the report should say about it

The four protocols

What the field tool asks for today on each system type. These are drafts I wrote; each one has an open box because you know what a maintenance stop on these actually involves.

6 **Straight cool — split system AC**

Six checks, eleven readings. The airside, refrigerant and electrical checks all carry measurements; condensate and controls are rated.

WHAT THE PROTOCOL CHECKS TODAY (6)

- M+T Airside performance return & supply dry bulb, return & supply static — delta-T and total static derived, judged against nameplate max ESP
 - M Refrigerant circuit suction and liquid saturation and line temps — superheat and subcooling derived, bands not set
 - M Condenser & heat rejection outdoor ambient — approach derived
 - M+T Electrical performance condenser and blower amps against nameplate RLA
 - R Condensate & drainage
 - R Controls, safeties & condition
- M+T measured with a target band · M measured, no target · R rating only, no reading

7 Heat pumps

Eight checks, twelve readings — the cooling protocol plus heating mode, defrost and backup heat. Charge and airflow do not change; what changes is that the system has a second job.

WHAT THE PROTOCOL CHECKS TODAY (8)

- M+T Airside performance
 - M Refrigerant circuit
 - M Outdoor coil & heat transfer
 - M Heating mode & defrost supply air in heating
 - R Backup heat & balance point
 - M+T Electrical performance
 - R Condensate & drainage
 - R Controls, safeties & condition
- M+T measured with a target band · M measured, no target · R rating only, no reading

8 Gas furnaces

Seven checks, five readings. Temperature rise is the one measurement with a real target available — the nameplate states it — and combustion analysis is not in the protocol at all yet, which is the obvious question.

WHAT THE PROTOCOL CHECKS TODAY (7)

- M Temperature rise return & supply dry bulb
- M+T Airflow & static return & supply static against nameplate max ESP
- R Burners, ignition & flame
- R Heat exchanger condition
- R Venting & combustion air

- M

Electrical & blower blower amps
- R

Safeties, controls & condition
- M+T measured with a target band · M measured, no target · R rating only, no reading

9 Mini-splits

Five checks, four readings — the thinnest protocol of the four, because static pressure has no meaning on a ductless head and most of what matters is blower wheel and coil condition.

- WHAT THE PROTOCOL CHECKS TODAY (5)
- M

Airside performance return & supply dry bulb
- M

Outdoor unit & heat transfer outdoor ambient
- M

Electrical performance running amps
- R

Condensate & drainage
- R

Controls, modes & condition
- M+T measured with a target band · M measured, no target · R rating only, no reading

Expected life and brand tiers

Same status as the appliance side: these figures are mine, they are drafts, and the app labels them as drafts on every report.

10 Expected service life — correct the draft

Texas duty cycle is the whole question. Change any cell that is wrong and leave the rest alone.

	LUXURY		MASS PREMIUM		MASS	
Split AC condenser	draft 18		draft 15		draft 12	
Heat pump	draft 16		draft 14		draft 11	
Gas furnace	draft 22		draft 18		draft 15	
Air handler / coil	draft 18		draft 15		draft 12	
Mini-split	draft 16		draft 13		draft 10	
Package unit	draft 16		draft 13		draft 10	

Write over anything that is wrong. A box left empty means the draft is fine.

What actually kills them here first — and whether a coil replacement should reset the clock

11 **Brand tiers**

Twenty brands are tiered in the draft and the tier drives the expected-life figure, so a brand in the wrong tier moves a customer's number.

Open floor

The three questions I cannot answer from a config file.

12 **What are we not asking that you would want on every stop?**

Anything you check by habit that is not in the four protocols above.

13 **Which of this is not worth the minutes?**

A maintenance stop has to fit in the day. If something above is padding, say so now rather than in six months.

14 **What does measureQuick do that you would miss?**

We use it and it is good. The intent is not to copy it but to keep what serves the customer, so it matters which parts of it you actually rely on — and which parts you work around.



Filled this in on a phone or laptop? Email the saved file to **cmayfield@wilsonappliance.com** and it goes straight into the shared sheet. Wrote on paper? Hand it in and it gets typed up once, by me.